NGUYEN VAN MUOI

AI & Operations Research Engineer | Neuro-Symbolic & Systems Specialist

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in **Neuro-Symbolic AI**, **Deep Reinforcement Learning (DRL)**, **Graph Neural Networks (GNN)**, and **Operations Research**. Experienced in architecting System 1 + System 2 document AI engines (VLM + Z3 SMT Solver), GNN-guided neural metaheuristics for VRPTW, and high-performance systems in Rust. Strong academic record featuring an **IELTS 7.5** score, **TDTU Entrance Scholarship**, coursework at **Chinese Culture University (Taiwan)**, and upcoming study at **VSB - Technical University of Ostrava (Czech Republic, 2027)**.

EDUCATION

Ton Duc Thang University (TDTU) — *Ho Chi Minh City, Vietnam* — 2022 – Present

Bachelor of Science in Computer Science (Undergraduate) — **TDTU Entrance Scholarship Recipient**

- **Thesis / Primary Research:** VRPTW Optimization via Hybrid GNN-DDQN+ALNS, advised by TS. Ho Thi Linh.

VSB – Technical University of Ostrava — *Ostrava, Czech Republic* — Upcoming 2027

Computer Science Study Program

Chinese Culture University (CCU) — *Taipei, Taiwan* — 2026

Completed Specialized Coursework in Artificial Intelligence & Computer Vision

TECHNICAL SKILLS & LANGUAGES

AI & Optimization: Neuro-Symbolic AI, Z3 SMT Solver, Qwen2.5-VL, Deep RL (DDQN), Graph Attention Networks (GAT), Multi-Task Learning (MTL), ConvNeXt-V2, ALNS Metaheuristics, PyTorch, Numba JIT.

Languages & Dev: Python, Rust, C++, C#, JavaScript / TypeScript, SQL, Git, Docker, Linux, FastAPI, TimescaleDB, Vite, Firebase, Vast.ai Cloud GPU.

Languages & Scores: **IELTS 7.5** (English), Basic Chinese (Mandarin), Vietnamese (Native).

KEY RESEARCH & PRIMARY OPEN-SOURCE PROJECTS

NeSy-DocAI: Neuro-Symbolic Document AI Research Engine *(github.com/Thundercok/nesy-docai)* — **[Python, PyTorch, Qwen2.5-VL, Z3 SMT Solver, Neuro-Symbolic AI]**

- Architected a System 1 + System 2 Neuro-Symbolic AI framework for auditable financial invoice parsing and formal verification.
- **System 1 (Visual Perception):** Deployed local Vision-Language Models (Qwen2.5-VL) to extract candidate lattices and layout bounding box coordinates.
- **System 2 (Symbolic Reasoning):** Integrated Z3 SMT Solver to formally verify accounting algebraic laws ($ ext{LineAmount}_i = ext{Qty}_i imes ext{UnitPrice}_i$) and perform constraint-guided error correction on OCR character mutations.

vrptw-neural-hybrid-optimizer (Project NAMI) *(github.com/Thundercok/vrptw-neural-hybrid-optimizer)* — **[Python, PyTorch, GNN/GAT, DDQN, ALNS, Numba JIT, FastAPI, Vite]**

- Engineered a GNN-guided Deep RL (DDQN) metaheuristic combining Graph Attention Networks (GAT) with ALNS for NP-hard vehicle routing problems with time windows.
- Developed GAT spatio-temporal node encoders (64-dim), Learned Acceptance Criterion (LAC), and a 6-mode search state switcher (`intensify`, `diversify`, `tw_rescue`, `pool_recombine`, `route_reduce`).
- Achieved **1.75% – 4.07% total travel distance reduction** over ALNS on 200-customer benchmarks and proved statistically significant vehicle count reduction edge on 400-customer scale ($p = 0.0078$, Wilcoxon test).

PG-MTAN: Physics-Guided Multi-Task Attention Network *(github.com/Thundercok/hcmc-traffic-analytics)* — **[PyTorch, ConvNeXt-V2, Multi-Task Learning, SAM 2, FastAPI, TimescaleDB]**

- Co-developed a multi-task deep neural framework (PG-MTAN) for real-time traffic density estimation and road flood monitoring via edge CCTV.
- Implemented an Orthogonal Gradient Disentanglement module ($\mathbf{g}_{ ext{flood}}^{ ext{proj}}$) to eliminate gradient interference between density estimation and water segmentation, achieving 54.0 FPS throughput on NVIDIA Jetson Orin hardware.

Spider-The-Web-Crawler *(github.com/Thundercok/Spider-The-Web-Crawler)* — **[Rust, Async/Tokio, High-Performance Systems]**

- Built a high-performance, asynchronous web crawling and scraping engine in Rust utilizing Tokio async runtime for concurrent pipeline throughput.

HONORS, COMPETITIONS & ORGANIZATION EXPERIENCE

TDTU Entrance Scholarship — *Ton Duc Thang University* — Scholarship Awardee

DataStorm Competition — *Data Science & AI Contest* — Participant

Vuon Uom Nha Nam — Vice Lead & Active Member

- Served as Vice Lead, directing organizational activities, team coordination, and project execution.

ITZone (Student IT Club at TDTU) — Department Lead, Host, Seminar Speaker & Mentor

- Led department operations, hosted technical seminars, and mentored student members in Computer Science and software engineering fundamentals.